CFAT 1.1.3

Mean, standard deviation and the normal distribution

1F, 1H, 1I · CAS and bound reference allowed

Name _____

/15

SECTION A · MULTIPLE CHOICE

Circle one answer. 6 marks

QUESTION 1 — 1 mark

For the following data set, the sample mean and standard deviation are closest to

91 56 67 34 65 89 76 99 33 21 23 54 34 32 76 67 84

A. mean = 58.9, standard deviation = 24.6
B. mean = 58.9, standard deviation = 25.4
C. mean = 25.4, standard deviation = 58.9
D. mean = 58.8, standard deviation = 24.6

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QUESTION 2 — 1 mark

A standardised score of $z = -2.4$ indicates an actual score is

A. less than 2 standard deviations greater than the mean.
B. more than 2 standard deviations greater than the mean.
C. equal to the mean.
D. more than 2 standard deviations less than the mean.

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QUESTION 3 — 1 mark

A data set has a mean of 52.0 and a standard deviation of 3.4.

Which of the following actual scores will give a standardised score of zero?

A. 0.0
B. 3.4
C. 52.0
D. 62.2

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QUESTION 4 — 1 mark

The time taken to *travel* between two regional cities is approximately normally distributed with a mean of 70 minutes and a standard deviation of 2 minutes.

The percentage of travel times that are between 66 minutes and 72 minutes is closest to

A. 2.5%
B. 34%
C. 68%
D. 81.5%
E. 95%

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QUESTION 5 — 1 mark

The pulse rates of a population of Year 12 students are approximately normally distributed with a mean of 69 beats per minute and a standard deviation of 4 beats per minute.

A student from this population has a standardised pulse rate of $z = -2.5$. This student's actual pulse rate is

A. 59 beats per minute.
B. 63 beats per minute.
C. 65 beats per minute.
D. 73 beats per minute.
E. 79 beats per minute.

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QUESTION 6 — 1 mark

The temperatures over a number of days in a city have a mean of 3.5 °C and a standard deviation of 3 °C. One of the temperatures is −4.3 °C.

The standardised value of this temperature is

A. −4.3
B. −2.6
C. 2.6
D. 3.5
E. 4.3

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SECTION B · SHORT ANSWER

Show your working. 9 marks

QUESTION 7

2 marks

In a large population of moths, the number of eggs per cluster is approximately normally distributed with a mean of 165 eggs and a standard deviation of 25 eggs.

Using the 68–95–99.7% rule, determine:

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a the percentage of clusters expected to contain more than 140 eggs.

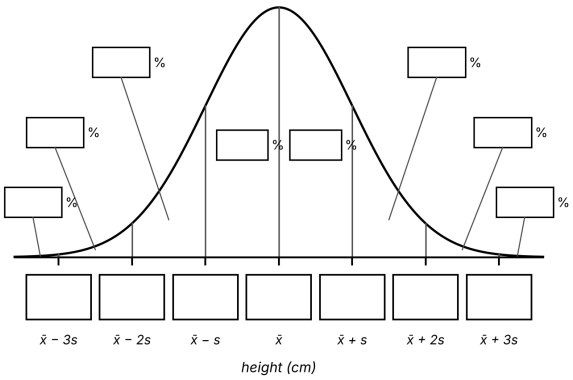
b the number of clusters expected to have less than 215 eggs in a sample of 1000 clusters.

QUESTION 8

5 marks

The heights of 2000 seedlings planted by Michelle in her garden were approximately normally distributed with a mean of 140 cm and a standard deviation of 12 cm.

a Fill in the blank boxes below



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b What percentage of the plants were between 128 cm and 140 cm?

c What percentage of the plants were between 104 cm and 116 cm?

d Michelle harvests any plants taller than 176 cm. Approximately how many plants will she need to harvest?

e The smallest 2.5% of seedlings need extra care. What height would the tallest of these seedlings be?

QUESTION 9

2 marks

The men's 100 m sprint results are approximately normally distributed. The mean times and standard deviations, in seconds, for 2017 to 2019 are:

2017: mean 10.43, SD 0.09
2018: mean 10.17, SD 0.23
2019: mean 9.91, SD 0.10

Kamil's best times were 10.55 s in 2017, 10.12 s in 2018 and 9.97 s in 2019.

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a Calculate Kamil's standardised score for each year?

b In which year did Kamil run fastest when compared against other athletes?